

Technical Documentation

Legal Document Customer Support RAG

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1 Introduction

What is RAG?

Retrieval-Augmented Generation (RAG) is a method that combines document retrieval with language generation to produce accurate and context-aware answers.

Why RAG?

1. Minimizes incorrect or hallucinated responses
2. Enhances answer accuracy using real data

Use Case

This system is designed to analyze legal documents and answer user queries efficiently.

2 System Architecture Explanation

Documents → Chunking → Embeddings →
Vector Store → Retrieval → Answer Generation
→ Output

3 Design Decisions

Chunk Size

A chunk size of 500 characters with 100-character overlap is used. This ensures that sufficient context is preserved while avoiding excessively large inputs that can slow down processing.

Embedding Strategy

Mini LM used for fast computation.

Retrieval Approach

Top-k similarity search is used to retrieve the most relevant chunks for a given query.

Prompt Design

- No hallucination
- Context-only answers

4 Work flow Explanation

Lang Graph Control

Lang Graph manages:

- Routing
- Processing
- Escalation

Node Responsibilities

1. Route Node → decides query category
2. Process Node → retrieves and answers
3. HITL Node → escalates when needed

5 Conditional Logic

Intent Detection

Simple rule-based logic is used.

Routing Decisions

1. If query is normal → generate answer
2. If query is unclear → escalate to human

6 HITL Implementation

Role

Handles queries where the system is not confident..

Benefits

1. Improves reliability
2. Prevents wrong answers

Limitations

1. Requires human intervention
2. Slower response for complex queries

7 Challenges & Trade-offs

Retrieval Accuracy vs Speed

More chunks improve accuracy but increase response time.

Chunk Size vs Quality

Larger chunks provide better context but slow down processing.

Cost vs Performance

Advanced models give better results but may increase cost.

8. Testing Strategy

Approach

The system is tested using multiple user queries related to the document.

Sample Queries

- “What is the termination clause?”
- “What are the penalties?”

9.Future Enhancements

1. Multi-document support
2. Web-based interface (Streamlit)
3. Better LLM integration
4. Feedback-based learning system
5. Cloud deployment

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